

MOU PAUL

Engineering leader focused on large scale infrastructure.
Proven track record of architecting pipelines for trillions of
training tokens, fighting spam at scale, and driving \$10M+
in infrastructure cost efficiencies

ppymou@gmail.com

paul.mou.dev

Technical Skills

Language Python Golang Typescript/Javascript Rust
Platform Kubernetes Docker Spark BigQuery BigTable Node.js gRPC
Infra Kafka Consul Vault Nginx Redis Sqlite
Library Ray Pytorch Rocksdb Electron Next.js React

Selected Projects

Building Flow Matching Models from Scratch
<https://paul.mou.dev/posts/2026-03-10-flow/>

Listening with LLM
<https://go.mou.dev/blog-llm>

Professional Experiences

Member of Technical Staff @ Microsoft AI, Redmond 2024/12-Present

- Architected petabyte-scale Spark/Ray data pipelines to process trillions of tokens from 3B+ unstructured PDFs for LLM pre-training, utilizing ablation studies to refine data-mixture strategies
- Engineered automated filtering and deduplication workflows using vector embeddings and LLM-as-a-Judge to curate high-fidelity training sets from raw, unstructured data

Software Engineer @ Roblox, San Mateo 2022/05-2024/12

- Led the evolution of Roblox's analytics ingestion infrastructure, scaling to trillions of events a day while delivering \$10M+ in annual cloud savings with datalake schematization
- Promoted DevOps best practices via code-based dashboard and alert generation, runbook documentation, capacity planning, and incident postmortems

Software Engineer @ Twitter, San Francisco 2018/06-2022/04

- Scaled smyte's anti-spam infrastructure 10x post-acquisition through rearchitecture and integration with Twitter infrastructure
- Hardened data privacy for GDPR compliance by designing a personal data anonymizer system for the anti-spam infrastructure

Software Engineer @ Smyte, San Francisco 2015/11-2018/06

- Core member of four engineers building and scaling Smyte's anti-spam engine protecting hundreds of millions of users
- Developed and maintained automation and monitoring for multiple kubernetes cluster with thousands of pods
- Contributed to high impact initiatives such as GDPR compliance and building first machine learning stack for abusive image detection

Educations

Udacity
Self-Driving Car Engineer I
Certificate

Stanford University
Computer Science
Master (withdrew)

University of Waterloo
System Design
Bachelor